

Number Play

CHAPTER 06

Learning Outcomes...

- Numbers Tell Us Things
- Picking Parity
- Magic Square : A Puzzle of Patterns
- The Virahāṅka Fibonacci Numbers
- Digits in Disguise (Cryptarithms)

Introduction

This chapter “Number Play” takes us beyond simple arithmetic. It invites you to explore how numbers behave, relate and sometimes surprise us through real world examples and fun mathematics puzzles. Whether it's understanding magic squares or recognizing pattern in number sequences and tricks to solve fun puzzles, number play helps us to explore interesting patterns, rules of numbers making maths fun and smart.

Numbers Tell Us Things

Numbers don't just show us “how much.” They can also help us to understand how things are lined up or how they are connected in a group or order. This means numbers can describe order (who comes first, second, third, etc.) and relationships (how things compare to others). Understanding these ideas helps us solve puzzles, understand patterns, and think logically.

For example: Look at this sequence of numbers:

2, 4, 6, 8, 10, 12

- Each number is 2 more than the previous one except 1st number.
- The rule here is “add 2.”

Now, let us have another example to understand order of numbers.

Imagine some books are placed in a line on a shelf from left to right. Each book has a number that tells that how many books to its left have more pages than that book.

- If a book's label is 0, it means no book to its left has more pages than this book.
- If a book's label is 2, it means two books to its left have more pages than this book.

Suppose we have 5 books with page counts:

Position on Shelf	Pages in Book	Number on Label (how many books to its left have more pages)
1	120	0
2	150	0

3	130	1
4	110	3
5	110	3

This shows a relationship between each book and the books before or left to it, helping us to understand the order and relative sizes without directly ranking everything.

ILLUSTRATIONS

1 Five children are holding ice cream cones with different numbers of scoops, but we don't know the exact number of scoops each one has.

Each child tells how many childrens before them in the line have more scoops than they have.

Here's what the children say (left to right):

Child A $\rightarrow$ 0

Child B $\rightarrow$ 1

Child C $\rightarrow$ 0

Child D $\rightarrow$ 2

Child E $\rightarrow$ 0

(i) Child B says 1, what does that mean?

(ii) Child C says 0, does C has more scoops than A and B?

(iii) Who has maximum number of scoops?

Soln.: (i) Child B says 1, meaning one person before him (Child A) has more scoops than B.

(ii) Child C says 0, so neither Child A nor B has more scoops than C.

(iii) Child E says 0, so no person before E has more scoops than him, as E is standing at last, so E has maximum number of scoops.

Picking Parity

Parity means whether a number is even or odd.

(i) An even number is divisible by 2 (For example; 2, 4, 6, 8).

(ii) An odd number leaves one as remainder when divided by 2 (For example; 1, 3, 5, 7)

Parity Rules

Now, let's understand what happens when we add or subtract even and odd numbers. These are called parity rules – and they help us to predict the result without calculating the full value.

Rule	Result	Example
Even + Even	Even	$4 + 6 = 10$
Odd + Odd	Even	$5 + 3 = 8$
Even + Odd	Odd	$2 + 5 = 7$

Let us look at an example : Riya has 4 packets of chocolates.

Each packet has an odd number of chocolates:

1, 3, 7, and 9

She wants to share all the chocolates equally between two friends – no chocolate should be left.

Let's add them:

$1 + 3 + 7 + 9 = 20$, this is an even number, which is divisible by 2. So chocolates can be divided between 2 friends by giving 10 chocolates to each to one friend.

Here, we get that sum of 4 odd numbers is even.

Now, suppose Riya has kept a packet of 9 chocolates with herself. Will she be able to share the remaining chocolates equally among two friends with no chocolate left.

Now, $1 + 3 + 7 = 11$, this is an odd number and an odd number can't be split into two equal whole parts — one chocolate will always be left.

Here, we have seen that when we add 4 odd numbers, we get an even number and when we add 3 odd numbers, we get an odd number. This gives rise to a few more rules of parity.

Few More Rules of Parity

Addition of Odd Numbers

- When we add odd numbers for an odd number of times, then the result will be an odd number.
- When we add odd numbers for an even number of times, then the result will be an even number.

For example; $3 + 7 + 9 + 11 = 30$ and $3 + 7 + 9 + 11 + 13 = 43$

Addition of Even Numbers

Adding even numbers for any number of times will result in an even number.

For example; $2 + 6 + 12 = 20$ and $4 + 16 + 12 + 18 = 50$

Sum of Two Consecutive Numbers

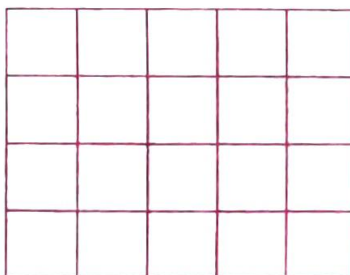
Whenever you pick two consecutive numbers — like 6 and 7 or 12 and 13 — one of them will always be even and the other will be odd. So, the sum of any two consecutive numbers will always be an odd.

For example; $6 + 7 = 13$ and $12 + 13 = 25$

Parity in Multiplication Grids

When we arrange squares in rows and columns — like tiles on a floor, we can calculate the total number of small squares by multiplying the number of rows and columns.

For example; a 4×5 grid has 20 small squares.



But if we only want to know whether the total number of squares is even or odd without multiplying the given number, then that can be done by using rules of parity under multiplication.

Instead of multiplying two large numbers, we can just check whether each number is odd or even, and use a simple rule to determine the parity of the resulting product.

Rule	Result	Example
Odd $\times$ Odd	Odd	$3 \times 5 = 15$
Even $\times$ Even	Even	$2 \times 4 = 10$
Even $\times$ Odd	Even	$2 \times 5 = 10$
Odd $\times$ Even	Even	$7 \times 6 = 42$

ILLUSTRATIONS

2 Raghav places 10 chairs in Row 1 and 9 chairs in Row 2, 8 chairs in Row 3, 7 chairs in Row 4, and 6 chairs in Row 5. Check the parity of total number of chairs.

Soln.: The arrangement of chairs is as follows:

- Row 1: 10 chairs
- Row 2: 9 chairs
- Row 3: 8 chairs
- Row 4: 7 chairs
- Row 5: 6 chairs

Total number of chairs = $10 + 9 + 8 + 7 + 6 = 40$

Since 40 is divisible by 2, it is an even number.

So, parity of total number of chairs is even.

3 Tina is packing juice boxes for a school picnic.

She has odd number of orange juice boxes (each costs ₹ 8); odd number of mango juice boxes (each costs ₹ 10) and even number of apple juice boxes (each costs ₹ 12)

She adds up the cost of all juice boxes and says the total is ₹ 200. Did Tina make a mistake? Explain why or why not.

Soln.: We know that,

- Odd $\times$ Even = Even
- Odd $\times$ Odd = Odd

• Even $\times$ Even = Even

Now, Orange juice: odd number $\times$ ₹ 8 (even)
→ Result is even

Mango juice: odd number $\times$ ₹ 10 (even) →
Result is even

Apple juice: even number $\times$ ₹ 12 (even) →
Result is even

So, all contributions are even.

Adding, even + even + even = even

Also, ₹ 200 is an even number. So, Tina's total is possible.

4 At a school function, rows of chairs are arranged neatly. The total number of chairs in each case is found by multiplying the number of rows and the number of chairs per row. Determine whether the total number of chairs is even or odd without actually multiplying?

(a) 25 rows with 18 chairs in each row.

(b) 37 rows with 47 chairs in each row.

Soln.: (a) 25 rows with 18 chairs in each row

Here, 25 is odd and 18 is even

Also, total number of chairs = 25×18

= Odd $\times$ Even = Even

So, the total number of chairs is even.

(b) 37 rows with 47 chairs in each row

Here, 37 is odd and 47 is also odd

Also, total number of chairs = 37×47

= Odd $\times$ Odd = Odd

So, the total number of chairs is odd.

Parity in Algebraic Expressions

In algebra, we often use expressions like $2n + 3$ to find the values based on a variable like n . But we can also use these expressions to predict whether the result will be even or odd without calculating the resulting value. This is called finding the parity of an expression.

For example ; Let's try to find parity of $5n + 1$ for different values of n :

n	$5n + 1$	Result
1	$5 \times 1 + 1 = 6$	Even
2	$5 \times 2 + 1 = 11$	Odd
3	$5 \times 3 + 1 = 16$	Even

As you can see, this expression can give both even and odd values. So, it does not always follow one parity. The result depends on the value of n .

Let us break down the expression to understand the parity of expression

When n is even	$5 \times n = \text{even}$	Even + 1 = odd
When n is odd	$5 \times n = \text{odd}$	Odd + 1 = even

So, when n is even, $5n + 1$ is odd and when n is odd, $5n + 1$ is even.

Expressions That Always Give Even Numbers

We know that when a number either even or odd is multiplied by an even number it will always result in an even number. Now for n be the variable, $n \times \text{even number} = \text{even number}$. So algebraic expressions like $2n$, $4n$, $6n$... will have even parity for any value of n .

Now, if we add or subtract any even number to an even number the result will be an even number.

So, $2n + 2$, $2n - 2$, $2n + 4$, $2n - 4$, $4n + 6$, $2n - 6$, are the algebraic expressions that has even parity.

So, we can say the expression $(n \times \text{even} \pm \text{even})$ has even parity.

Note : If an expression has 2 as a common factor in all terms, it usually gives an even number.

Expressions That Always Give Odd Numbers

When a variable n is multiplied by an even number, then we get the expression $2n$, $4n$ $6n$ and so on which has even parity. Now if we add or subtract any odd number to it we will get an expression having odd parity.

So, $2n + 1$, $2n - 1$, $2n + 3$, $2n - 3$, $4n + 5$, $4n - 5$ are the algebraic expressions that has odd parity.

So, we can say the expression $(n \times \text{even} \pm \text{odd})$ has odd parity.

Finding the n^{th} Even or Odd Number

n^{th} Even Numbers

As, 1st even number = $2 = 2 \times 1$; 2nd even number = $4 = 2 \times 2$; 3rd even number = $6 = 2 \times 3$

We observe that the smallest even number is 2 and each even number is 2 times its position.

So, n^{th} even Number = $2 \times n$

For example ; 10^{th} even number $= 2 \times 10 = 20$.

n^{th} Odd Numbers

1^{st} odd number $= 1 = 2 \times 1 - 1$; 2^{nd} odd number $= 3 = 2 \times 2 - 1$; 3^{rd} odd number $= 5 = 2 \times 3 - 1$

We observe that the smallest odd number is 1 and every odd numbers are 1 less than even numbers.

So, the n^{th} odd Number $= 2 \times n - 1$

For example; 10^{th} odd number $= 2 \times 10 - 1 = 19$

ILLUSTRATIONS

5 Write two expressions, which always have odd parity, and justify your answer.

Soln.: Let n be a variable then,

(i) the term $2n$ gives us an even number (because 2 times any whole number is even).

(ii) adding 1 to an even number always gives an odd number.

So, $2n + 1$ will always give an odd number.

Justification :

If $n = 3$, then $2 \times 3 + 1 = 6 + 1 = 7$ (odd)

If $n = 4$, then $2 \times 4 + 1 = 8 + 1 = 9$ (odd)

Now, for another expression let, k be a variable then,

(i) the term $4k$ is always even (4 times any

whole number is even).

(ii) subtracting 1 from an even number gives an odd number.

So, $4k - 1$ always gives an odd number.

Justification :

If $k = 2$, then $4 \times 2 - 1 = 8 - 1 = 7$ (odd)

If $k = 3$, then $4 \times 3 - 1 = 12 - 1 = 11$ (odd)

6 Find the 22^{nd} , 40^{th} and 88^{th} odd numbers.

Soln.: We know that n^{th} odd number $= 2n - 1$

22^{nd} odd number $= 2 \times 22 - 1 = 44 - 1 = 43$

40^{th} odd number $= 2 \times 40 - 1 = 80 - 1 = 79$

88^{th} odd number $= 2 \times 88 - 1 = 176 - 1 = 175$

Magic Square : A Puzzle of Patterns

Magic Square

A magic square is a square grid (usually of integers), where the numbers are arranged such that the sum of the number in each row, column and both main diagonals is the same. This common sum is called the magic sum or magic constant.

For example; Consider the 3×3 grid :

2	7	6
9	5	1
4	3	8

Now, let's add the numbers in each row, each column and main diagonals one by one.

Rows : $2 + 7 + 6 = 15$; $9 + 5 + 1 = 15$; $4 + 3 + 8 = 15$

Columns : $2 + 9 + 4 = 15$; $7 + 5 + 3 = 15$; $6 + 1 + 8 = 15$

Diagonals : $2 + 5 + 8 = 15$; $6 + 5 + 4 = 15$

Each row, each column and main diagonals adds up to the same number 15. So, this is a magic square, and 15 is the magic sum.

Note : Since, there are 3 rows and the sum of digit from 1 to 9 is 45 and $45 \div 3$ is 15, that means the magic sum for a 3×3 magic square using numbers 1 to 9 is always 15.

Make Your Own 3×3 Magic Square

Forming a magic square might look tricky at first, but we can create it by just following the simple method to fill a 3×3 grid using numbers from 1 to 9 (without repeating any).

Firstly, we need to fill a 3×3 grid by choosing numbers from 1 to 9, where sum of each row, each column and main diagonals is 15.

Step 1 : Always put 5 in the centre.

	5	

Step 2 : Put even numbers in the corner : having either 2 and 8 or 4 and 6 in the same unique diagonal. This can be done in 8 way.

2		4
	5	
6		8

2		6
	5	
4		8

4		2
	5	
8		6

4		8
	5	
2		6

6		8
	5	
2		4

6		2
	5	
8		4

8		6
	5	
4		2

8		4
	5	
6		2

Step 3 : As, each row, each column and main diagonals add to 15 so, remaining places can be filled using the digits 1, 3, 7, 9.

2	9	4
7	5	3
6	1	8

2	7	6
9	5	1
4	3	8

4	9	2
3	5	7
8	1	6

4	3	8
9	5	1
2	7	6

6	1	8
7	5	3
2	9	4

6	7	2
1	5	9
8	3	4

8	1	6
3	5	7
4	9	2

8	3	4
1	5	9
6	7	2

► There are exactly 8 unique ways to make a 3×3 magic square using numbers 1 to 9.

► There is only one basic 3×3 magic square using numbers 1 to 9, and all other versions are just rotations or reflections of it.

COMPETITION WINDOW



Generalising a 3×3 Magic Square

When we generalise a magic square, we try to find a mathematical rule that helps us to construct a magic square, no matter what number is placed at the centre.

We can express all the other numbers in the grid in terms of the central number.

2	9	4
7	5	3
6	1	8

Here's a magic square with numbers from 1 to 9:

In this square, the centre number is 5, and all rows, columns, and diagonals add up to 15.

Let's call the centre number n .

Now, let's describe each number in relation to the centre number, $n = 5$.

Converting Numbers into Expressions:

We rewrite each number in terms of how much more or less it is than 5:

So, the generalised magic square becomes:

$n - 3$	$n + 4$	$n - 1$
$n + 2$	n	$n - 2$
$n + 1$	$n - 4$	$n + 3$

Note : (i) In generalised magic square : All rows, columns, and diagonals will add up to the same expression i.e., n .

(ii) Magic Sum = $3n$ i.e., Every row, column, and diagonal add up to $3 \times$ (center value).

(iii) The generalised form of a 3×3 magic square depends on the selection of specific magic square pattern used as the base.

ILLUSTRATIONS

7 Using the above given generalised form, create a magic square where the center is 7.

Soln.: Let us use the general form of 3×3 magic square given below.

$n - 3$	$n + 4$	$n - 1$
$n + 2$	n	$n - 2$
$n + 1$	$n - 4$	$n + 3$

On substituting $n = 7$ we get,

4	11	6
9	7	5
8	3	10

Check Magic Sum:

$$7 \times 3 = 21$$

Each row, column and diagonal add up to 21.

8 You are given that in a 3×3 magic square, the top row is:

$x - 3$	$x + 4$	$x - 1$
---------	---------	---------

The magic sum is 90.

Find the value of the center number x , and complete the entire magic square.

Soln.: Each row sum = $3x$

$$\text{So, } 3x = 90 \Rightarrow x = 30$$

Substitute into the 3×3 general square:

$x - 3 = 27$	$n + 4 = 34$	$x - 1 = 29$
$x + 2 = 32$	$x = 30$	$x - 2 = 28$
$x + 1 = 31$	$x - 4 = 26$	$x + 3 = 33$

So, required magic square is

27	34	29
32	30	28
31	26	33

The First Ever Magic Square

The Chautisā Yantra (meaning "34 Device") is the first recorded 4×4 magic square found in India, carved on the walls of the Pārshvanath Jain Temple in Khajuraho, dating back to the 10th century CE.

7	12	1	14
2	13	8	11
16	3	10	5
9	6	15	4

Every row, column, and diagonal adds up to 34 — that's why it's called Chautīsā, which means 34 in Hindi. Not just rows and column but many groups of four numbers (like corners, (2×2) squares) and broken diagonals) also add up to 34.

India's Rich History of Magic Square

Magic squares have a long and fascinating history in India, where they were not only mathematical curiosities but also held deep spiritual and cultural significance.

- The Palani Murugan Temple in Tamil Nadu has an ancient 3×3 magic square carved on one of its pillar from the 8th century CE.
- Navagraha Yantras and Kuber Yantra are also arranged in magic square formats, symbolizing planetary energies and prosperity.

The Virahāṅka Fibonacci Numbers

The Virahāṅka sequence, known today as the Fibonacci sequence, actually has ancient Indian origins rooted in the world of poetry — not mathematics.

In ancient Indian poetry, especially in Sanskrit and Prakrit languages, each syllable (a sound unit) was either : a short syllable (called *laghu*) i.e., 1 beat and a long syllable (called *guru*) i.e., 2 beats.

Poets used to compose verses using combinations of these syllables.

Then mostly asked question is “In how many different ways we can create a rhythmic line that's exactly 8 beats long using just short and long syllables?”

Here are some examples:

(i) $1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$

(ii) $2 + 2 + 2 + 2$

(iii) $1 + 2 + 2 + 1 + 2$

and so on.

Each different combination is called a rhythmic pattern. This question led to a mathematical idea “How many total rhythmic combinations (patterns) are possible for any given number of beats?”

Because it is not easy to count number of patterns for each and every beat.

For example ;

Beats (n)	Rhythms (way to arrange 1 s and 2 s)
1	1
2	$2(1 + 1, 2)$
3	$3(1 + 1 + 1, 2 + 1, 1 + 2)$
4	$5(1 + 1 + 1 + 1, 1 + 1 + 2, 1 + 2 + 1, 2 + 1 + 1, 2 + 2)$

and so, on

This problem led the scholar Virahāṅka, around 700 CE, to discover a special number pattern. He realized that to find the number of patterns for 5 beats, we could simply add:

- (i) the number of patterns for 4 beats, and
- (ii) the number of patterns for 3 beats.

This simple idea led to a very famous number sequence :

1, 2, 3, 5, 8, 13, 21, 34, 55, ... where

"Each number is the sum of the two numbers before it."

This is called the Virahāṅka sequence in India (and also known as the Fibonacci sequence in the West).

➡ The number of rhythms for n beats = rhythms for $(n-1)$ beats + rhythms for $(n-2)$ beats.

COMPETITION WINDOW



Parity of Virahāṅka Numbers

Let's write down the parity (even or odd) for each number:

Term	Number	Parity
1 st	1	Odd
2 nd	2	Even
3 rd	3	Odd
4 th	5	Odd
5 th	8	Even
6 th	13	Odd
7 th	21	Odd
8 th	34	Even
9 th	55	Odd
10 th	89	Odd

On noticing the pattern, we observe that it starts with odd, then even and after that, we have 2 odds following an even number and the pattern goes on like this.

Virahāṅka Sequence in Nature

Many flowers have petals that match numbers from the Virahāṅka sequence.

Flower Name	Number of Petals	Is it a Virahāṅka Number?
Lily	3	Yes

Buttercup	5	Yes
Black-eyed susan	13 or 21 or 34	Yes

Virahāṅka Numbers in Art and Science

These numbers also appear in :

- Music rhythms (like in drumming or classical poetry)
- Spirals in pinecones, pineapples, and sunflowers
- Architectural designs and art compositions
- Computer algorithms and data structures
- DNA molecules and biological growth patterns.

ILLUSTRATIONS

9 In the given 4×4 magic square select any five different 2×2 blocks within the square whose total sum is 34.

7	12	1	14
2	13	8	11
16	3	10	5
9	6	15	4

Soln.: Let us try the corner blocks

Block 1 :

7	12
2	13

$$\text{Sum} = 7 + 12 + 2 + 13 = 34$$

Block 2 :

16	3
9	6

$$\text{Sum} = 16 + 3 + 9 + 6 = 34$$

Block 3 :

1	14
8	11

$$\text{Sum} = 1 + 14 + 8 + 11 = 34$$

Block 4 :

10	5
15	4

$$\text{Sum} = 10 + 5 + 15 + 4 = 34$$

Now for fifth block let us take middle block

Block 5 :

13	8
----	---

3	10
---	----

$$\text{Sum} = 13 + 8 + 3 + 10 = 34$$

So, here we get 5, 2×2 blocks whose sum is 34.

10 How many different rhythmic patterns can be created using short (1 beat) and long (2 beat) syllables for 9 beats?

Soln.:

Beats (n)	Rhythmic Patterns
1	$1 \rightarrow (1)$
2	$2 \rightarrow (1 + 1), (2)$
3	$3 \rightarrow (1 + 1 + 1), (1 + 2), (2 + 1)$
4	$5 \rightarrow \text{pattern for 3 beats} + \text{pattern for 2 beats} = 3 + 2$
5	$8 \rightarrow 5 + 3$
6	$13 \rightarrow 8 + 5$
7	$21 \rightarrow 13 + 8$
8	$34 \rightarrow 21 + 13$
9	$55 \rightarrow 34 + 21$

So, there are 55 different rhythmic patterns possible using short (1 beat) and long (2 beat) syllables to make exactly 9 beats.

11 Which of the following flowers follow the Virahāṅka pattern?

- Sunflower with 34 petals
- Lotus with 8 petals
- Rose with 17 petals
- Daisy with 21 petals

Soln.: We have, Virahāṅka sequence: 1, 2, 3, 5, 8, 13, 21, 34, 55...

(i) Sunflower having 34 petals follow the Virahāṅka pattern as it is a Virahāṅka number.

(ii) Lotus having 8 petals follow the Virahāṅka pattern as it is a Virahāṅka number.

(iii) Rose having 17 petals does not follow the Virahāṅka pattern as it is not a Virahāṅka number.

(iv) Daisy having 21 petals follow the Virahāṅka pattern as it is a Virahāṅka number.

All the given flower follows the pattern except Rose.

Digits in Disguise (Cryptarithm)

Digit in Disguise means that a digit (number) is hidden behind a letter in a math problem. It means that instead of seeing regular numbers like 3 or 7, we see letters like A, B, C, X or Y. But each letter stands for a specific digit from 0 to 9 and our job is to figure out what digit that letter is hiding. Cryptarithm is a type of math puzzle where letters are used instead of digits in full addition, subtraction, multiplication, or division problems. The goal is the same i.e., to find which digit each letter represents.

Consider the given combination

$$\begin{array}{r} \text{For example:} \quad M \\ + M \\ + M \\ \hline XM \end{array}$$

Here, a one-digit number M, when added to itself three times, gives a two-digit number XM, where the units digit is again M.

Let us try to find the value of X and M.

We know $M + M + M = 3M$

So, $3M = XM$ (a number ending in M)

Try with different values of M :

- $M = 1 \rightarrow 3 \times 1 = 3 \rightarrow$ not a two-digit number
- $M = 2 \rightarrow 6 \rightarrow$ not a two-digit number
- $M = 3 \rightarrow 9 \rightarrow$ not a two-digit number
- $M = 4 \rightarrow 12, \rightarrow$ two-digit number but unit digit is not 4
- $M = 5 \rightarrow 15 \rightarrow$ two-digit number, also units digit is 5

So, $M = 5$, $X = 1$ and $XM = 15$

ILLUSTRATIONS

12 Solve the following cryptarithms :

$$\begin{array}{r} 3 \ 1 \ A \\ + 1 \ A \ 3 \\ \hline 5 \ 0 \ 1 \end{array}$$

Soln.: We have,

$$\begin{array}{r} 3 \ 1 \ A \\ + 1 \ A \ 3 \\ \hline 5 \ 0 \ 1 \end{array}$$

Here, we have to find the value of A, which can take values from 0 to 9.

As, A take values from 0 to 9. Therefore, $A + 3$ can take values from 3 to 12. Since, digit at the units place of the sum of two digits A and 3 is 1. Therefore, $A + 3$ is a number between 3 and 12 whose units digit is 1. Clearly, such a number between 3 and 12 is 11.

$$\therefore A + 3 = 11 \Rightarrow A = 8$$

This value of A satisfies the addition in tens and hundreds columns.

Hence, $A = 8$.

13 Find the value of A , B and C .

$$\begin{array}{r} 2 \ A \ 6 \ 2 \\ + \ A \ 2 \ 4 \ A \\ \hline B \ C \ C \ B \ 1 \end{array}$$

Soln.: ' A ' may be any number from 0 to 9.

But it cannot be zero as $2 + A$ is given to be 1. Also ' A ' cannot be any number upto 8 as the sum of 2 and A must have 1 at the unit place. This can happen only when $A = 9$ because $2 + 9 = 11$, which has 1 at the unit place and 10 is carried to the next digit.

Moving on to the next column on the left, we find that 6 tens + 4 tens + 1 ten (carried) gives 11 tens. Hence, B has to be one and one hundred is carried to next column.

In hundreds place $A = 9$, so $900 + 200 + 100$ (carried) makes it 12 hundreds. So, 2 from 12 comes in place of C . Thus, C has to be 2.

$$\begin{array}{r} 2 \ 9 \ 6 \ 2 \\ + \ 9 \ 2 \ 4 \ 9 \\ \hline 1 \ 2 \ 2 \ 1 \ 1 \end{array}$$

Hence, $A = 9$, $B = 1$ and $C = 2$

NUMBER PLAY

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Picking Parity

Parity means whether a number is even or odd.

- (i) An even number is divisible by 2 (For example, 2, 4, 6, 8).
- (ii) An odd number leaves one as remainder when divided by 2 (For example, 1, 3, 5, 7)

Parity in Algebraic Expressions

In algebra, we often use expressions like $2n + 3$ to find the values based on a variable like n . But we can also use these expressions to predict whether the result will be even or odd without calculating the resulting value. This is called finding the parity of an expression.

Magic Square

A magic square is a square grid (usually of integers), where the numbers are arranged such that the sum of the numbers in each row, column and both main diagonals is the same. This common sum is called the magic sum or magic constant.

The first ever Magic Squares

The Chautisā Yantra (meaning "34 Device") is the first recorded 4×4 magic square found in India, carved on the walls of the Pārśvanath Jain Temple in Khajuraho, dating back to the 10th century CE.

7	12	1	14
2	13	8	11
16	3	10	5
9	6	15	4

Every row, column, and diagonal adds up to 34 – that's why it's called Chautisā, which means 34 in Hindi. Not just rows and columns but many groups of four numbers (like corners, 2×2 squares) and broken diagonals) also add up to 34.

The Virahāṅka Fibonacci numbers

The Virahāṅka sequence, known today as the Fibonacci sequence, actually has ancient Indian origins rooted in the world of poetry – not mathematics.

a very famous number sequence :

1, 2, 3, 5, 8, 13, 21, 34, 55, ... where

"Each number is the sum of the two numbers before it."

Digits in Disguise (Cryptarithm)

Digit in Disguise means that a digit (number) is hidden behind a letter in a math problem.

It means that instead of seeing regular numbers like 3 or 7, we see letters like A, B, C, X or Y. But each letter stands for a specific digit from 0 to 9 and our job is to figure out what digit that letter is hiding. Cryptarithm is a type of math puzzle where letters are used instead of digits in full addition, subtraction, multiplication, or division problems. The goal is the same i.e., to find which digit each letter represents.

Solved Examples

1. Each time the doorbell button is pressed, the bell rings if it was silent, or stops ringing if it was already ringing. Initially, the doorbell is silent. Rina presses the doorbell button 125 times. After these presses, will the doorbell be ringing or silent? Explain your answer.

Soln.: Each press toggles the doorbell:

Press 1: Rings (ON)

Press 2: Silent (OFF)

Press 3: Rings (ON)

... and so on.

Since, pressing toggles the state, the doorbell's final state depends on whether the number of presses is odd or even:

Odd number of presses → The doorbell is ON (ringing).

Even number of presses → The doorbell is OFF (silent).

Given 125 is odd, the doorbell will be ringing (ON) on 125th press.

2. Find the next two numbers in the sequence: 3, 6, 12, 24, 48, ...

What is the 8th term in the sequence? Also, explain the pattern.

Soln.: Each number doubles the previous one:

$$3 \times 2 = 6$$

$$6 \times 2 = 12$$

$$12 \times 2 = 24$$

$$24 \times 2 = 48$$

So, the next two numbers are

$$48 \times 2 = 96$$

$$96 \times 2 = 192$$

We have, term 6 = 96

Term 7 = 192

So, term 8 = $192 \times 2 = 384$

3. Does the expression $5x + 2$ always produce even numbers? Explain why or why not?

Soln.: To determine whether the expression $5x + 2$ always gives even numbers, we need to analyze it for different values (odd or even) of x . Since, 5 is an odd number, the product $5x$ will have the same parity as x .

(i) If x is even, then $5x$ is even (odd $\times$ even = even).

(ii) If x is odd, then $5x$ is odd (odd $\times$ odd = odd). Now, adding 2 (which is even) to $5x$ affects parity as follows :

(i) Even + Even = Even

(ii) Odd + Even = Odd

For $x = 1$ (odd) :

$$5 \times 1 + 2 = 5 + 2 = 7$$

⇒ Odd

For $x = 2$ (even) :

$$5 \times 2 + 2 = 10 + 2 = 12$$

⇒ Even

So, the expression $5x + 2$ produces even numbers when x is even and odd numbers when x is odd. Therefore, $5x + 2$ does not always produce even numbers.

4. Make a 3×3 magic square with 12 as the magic sum using the given generalised form.

$n - 3$	$n + 4$	$n - 1$
$n + 2$	n	$n - 2$
$n + 1$	$n - 4$	$n + 3$

Soln.: The magic sum (S) of a 3×3 magic square can be expressed as:

$S = 3 \times$ (centre number), because the centre number say n is counted in a row, a column, and both diagonals.

So, for our magic sum $S = 12$

$$12 = 3 \times n \Rightarrow n = 4$$

Hence, the center cell of the square is 4.

Now, by using general 3×3 magic square

$n - 3$	$n + 4$	$n - 1$
$n + 2$	n	$n - 2$
$n + 1$	$n - 4$	$n + 3$

We get a magic square :

1	8	3
6	4	2
5	0	7

Here, sum of all rows, columns and diagonals is 12.

5. A parking lot is designed with an odd number of rows and an odd number of parking spots in each row. The owner wants to make sure the total number of parking spots is even. Can the owner achieve this by adding just one row or one parking spot per row? Explain your answer with reason.

Soln.: Number of rows = odd number (let's call it R)

Number of parking spots in each row = odd number (let's call it S)

The total number of parking spots is calculated by multiplying the number of rows by the number of spots per row:

$$\text{Total spots} = R \times S$$

Since, both R and S are odd, and $\text{Odd} \times \text{Odd} = \text{Odd}$. This means the total number of parking spots is odd.

If the owner adds one row, the new number of rows becomes $R + 1$.

Since, R is odd, $R + 1$ even and number of spots per row remains the same (odd number S).

$$\begin{aligned} \text{New total number of spots} &= (R + 1) \times S \\ &= \text{Even} \times \text{Odd} = \text{Even} \end{aligned}$$

So, adding one row changes the total number of spots to an even number.

Now, if the owner adds one parking spot to each row, the number of spots per row becomes $S + 1$

Since, S is odd, $S + 1$ is even and number of rows remains the same (odd number R).

$$\begin{aligned} \text{New total number of spots} &= R \times (S + 1) \\ &= \text{Odd} \times \text{Even} = \text{Even} \end{aligned}$$

So, adding one parking spot per row also changes the total number of spots to an even number. Hence, there are two ways that will change the total number of spots to an even number.

6. You have a multiplication grid where the number of rows is a two-digit odd number, and the number of columns is a two-digit even number. Without multiplying, how do you find whether the total number of cells is odd or even? Give two examples to support your answer.

Soln.: We have, Rows = a two-digit odd number (e.g., 11, 15, 39, etc.)

And columns = a two-digit even number (e.g., 12, 28, 64, etc.)

Let R denotes the number of rows (odd) and C denotes the number of columns (even).

Then, total number of cells in the grid = $R \times C$

Since, $\text{Odd} \times \text{Even} = \text{Even}$.

So, the total number of cells will always be even.

For example; (i) Let number of rows = 15 (odd), and columns = 12 (even).

$$\text{Total cells} = 15 \times 12 = 180 = \text{Even}$$

(ii) Let number of rows = 39 (odd),

and columns = 28 (even)

$$\text{Total cells} = 39 \times 28 = 1092 = \text{Even}$$

7. Does diamond patterns in a 4×4 Chautisi Yantra add up to 34?

7	12	1	14
2	13	8	11
16	3	10	5
9	6	15	4

Soln.: Let's find the sum of the diamond-shaped group (12, 8, 3, 2).

$12 + 8 + 2 + 3 = 25$ so, the inner diamond adds up to 25, not 34. Hence, we can say that not every group forms 34, only special patterns can do.

8. Can you find a group of four numbers not in the same row/column of Chautisi Yantra that still add up to 34? Justify

Soln.: Let us take 4 numbers from different rows of Chautisā Yantra. $12(\text{from Row1}) + 8(\text{Row2}) + 10(\text{Row3}) + 4(\text{Row4}) = 12 + 8 + 10 + 4 = 34$

So, a random combination can also form magic sum — that's part of the beauty of this square.

9. Create all 8 rhythmic patterns for 5 beats.

Soln.: The following are 8 rhythmic patterns for 5 beats.

1. $1 + 1 + 1 + 1 + 1$
2. $1 + 1 + 1 + 2$
3. $1 + 1 + 2 + 1$
4. $1 + 2 + 1 + 1$
5. $2 + 1 + 1 + 1$
6. $1 + 2 + 2$
7. $2 + 1 + 2$
8. $2 + 2 + 1$

10. Radha wants to compose a verse that is exactly 7 beats long. She begins with a long syllable (2 beats). How many rhythmic patterns are possible now?

Soln.: If she starts with 2 beats, 5 beats remain.

Remaining rhythms = Rhythms for 5 beats = 8.

So, 8 patterns are possible.

11. If a verse of 10 beats has 89 rhythmic patterns, then how many rhythmic will a verse of 11 beats have?

Soln.: We also need the number of patterns for 9 beats to proceed.

From the earlier known Virahāṅka (Fibonacci) sequence:

Beats (n)	Rhythmic Patterns
1	1
2	2
3	3
4	5
5	8
6	13
7	21
8	34
9	55
10	89

Now, by using the rule we get,

Patterns for 11 beats = Patterns for 10 beats

+ Patterns for 9 beats = $89 + 55 = 144$.

12. The 9th and 10th Virahāṅka numbers are 55 and 89. What is the parity of the 11th number?

Soln.: We have, 9th number = 55, which is odd and 10th number = 89, which is also odd.

Now, 11th number of sequence = $55 + 89 = 144$ and parity of 144 is even

∴ The 11th term is even.

13. Below is a 3×3 grid.

	C1	C2	C3	
R1				→ e
R2				→ o
R3				→ e
	↓ o	↓ e	↓ e	

Each row and column shows the parity of the sum:

- 'e' means the sum of the row or column must be even

- 'o' means the sum must be odd

You must fill the grid with 5 odd numbers and 4 even numbers, using digits from 1 to 9 (no repeats), so that the sum of each row and column matches the given parity.

Soln.: Odd numbers are 1, 3, 5, 7, 9 and even numbers are 2, 4, 6, 8.

We need to choose the numbers in such way, we get the following parity.

Rows: R1 → even, R2 → odd, R3 → even

Columns: C1 → odd, C2 → even, C3 → even

Let us choose R1 as 1, 2, 3, R2 as 6, 7, 8 and R3 as 4, 5, 9. Now, let us check the parity :

1	2	3	→ $1 + 2 + 3 = 6 = \text{even}$
6	7	8	→ $6 + 7 + 8 = 21 = \text{odd}$
4	5	9	→ $4 + 5 + 9 = 18 = \text{even}$
↓	↓	↓	
$1 + 6 + 4 =$ $11 = \text{odd}$		$3 + 8 + 9 =$ $20 = \text{even}$	
	↓		
	$2 + 7 + 5 =$ $14 = \text{even}$		

Here, we get the required parity.

14. If $A \times 3 = BA$, then find the possible values of A and B .

Soln.: Let us take $A = 1$ then, $1 \times 3 = 3$, this does not give a two digit number ending with A .

Now, take $A = 4$.

So, $4 \times 3 = 12$ which ends in $2 \neq A$

Let $A = 5$ then $3 \times 5 = 15$, Ends in $5 = A$

So, $A = 5$, $BA = 15$

$\Rightarrow B = 1$

Only in case of 5 we get the unit digit same.

15. A portion of the Virahāṅka sequence is missing. Fill the blank spaces :

610, ?, 1597, ?, 4181 ...

Soln.: We move forward by adding previous two terms. If 610 is before the first blank then,

First blank $\Rightarrow 610 + ? = 1597$

So, missing number $= 1597 - 610 = 987$

Now, we have: 610, 987, 1597

Next blank $= 1597 + 987 = 2584$

$\therefore$ Required Sequence is 610, 987, 1597, 2584, 4181, ...

NCERT Section

Figure it Out

(Page 128)

1. Arrange the stick figure cut-outs given at the end of the book, or draw a height arrangement such that the sequence reads:

- (a) 0, 1, 1, 2, 4, 1, 5
- (b) 0, 0, 0, 0, 0, 0, 0
- (c) 0, 1, 2, 3, 4, 5, 6
- (d) 0, 1, 0, 1, 0, 1, 0
- (e) 0, 1, 1, 1, 1, 1, 1
- (f) 0, 0, 0, 3, 3, 3, 3

2. For each of the statements given below, think and identify if it is Always True, Only Sometimes True, or Never True. Share your reasoning.

- (a) If a person says '0', then they are the tallest in the group.
- (b) If a person is the tallest, then their number is '0'.
- (c) The first person's number is '0'.
- (d) If a person is not first or last in line (i.e., if they are standing somewhere in between), then they cannot say '0'.
- (e) The person who calls out the largest number is the shortest.
- (f) What is the largest number possible in a group of 8 people?

Figure it Out

(Page 131)

1. Using your understanding of the pictorial representation of odd and even numbers, find out the parity of the following sums:

- (a) Sum of 2 even numbers and 2 odd numbers (e.g., even + even + odd + odd)
- (b) Sum of 2 odd numbers and 3 even numbers
- (c) Sum of 5 even numbers
- (d) Sum of 8 odd numbers

2. Lakpa has an odd number of ₹ 1 coins, an odd number of ₹ 5 coins, and an even number of ₹ 10 coins in his piggy bank. He calculated the total and got ₹ 205. Did he make a mistake? If he did, explain why. If he didn't, how many coins of each type could he have?

3. We know that:

- (a) even + even = even
- (b) odd + odd = even
- (c) even + odd = odd

Similarly, find out the parity for the scenarios below:

- (d) even - even = _____
- (e) odd - odd = _____
- (f) even - odd = _____
- (g) odd - even = _____

Figure it Out

(Page 136)

- How many different magic squares can be made using numbers 1-9?
- Create a magic square using numbers 2-10. What strategy would you use for this? Compare it with the magic squares made using 1-9.

- Take a magic square, and
 - increase each number by 1
 - double each number

In each case, is the resulting grid also a magic square? How do the magic sums change in each case?

- What other operations can be performed on a magic square to yield another magic square?
- Discuss ways of creating a magic square using any set of 9 consecutive numbers (like 2-10, 3-11, 9-17, etc.).

Figure it Out

(Page 137)

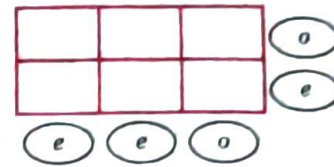
- Using this generalised form, find a magic square if the centre number is 25.
- What is the expression obtained by adding the 3 terms of any row, column or diagonal?
- Write the result obtained by
 - adding 1 to every term in the generalised form.
 - doubling every term in the generalised form.
- Create a magic square whose magic sum is 60.
- Is it possible to get a magic square by filling nine non-consecutive numbers?

Figure it Out

(Page 143 & 144)

- A light bulb is ON. Dorjee toggles its switch 77 times. Will the bulb be on or off? Why?
- Liswini has a large old encyclopaedia. When she opened it, several loose pages fell out of it. She counted 50 sheets in total, each printed on both sides. Can the sum of the page numbers of the loose sheets be 6000? Why or why not?
- Here is a 2×3 grid. For each row and column, the parity of the sum is written in the circle; 'e' for even and 'o' for odd. Fill the 6 boxes with 3 odd numbers ('o') and 3 even

numbers ('e') to satisfy the parity of the row and column sums.



- Make a 3×3 magic square with 0 as the magic sum. All numbers cannot be zero. Use negative numbers, as needed.
- Fill in the following blanks with 'odd' or 'even':
 - Sum of an odd number of even numbers is _____
 - Sum of an even number of odd numbers is _____
 - Sum of an even number of even numbers is _____
 - Sum of an odd number of odd numbers is _____
- What is the parity of the sum of numbers from 1 to 100?
- Two consecutive numbers in the Virahāṅka sequence are 987 and 1597. What are the next 2 numbers in the sequence? What are the previous 2 numbers in the sequence?
- Angaan wants to climb an 8-step staircase. His playful rule is that he can take either 1 step or 2 steps at a time. For example, one of his paths is 1, 2, 2, 1, 2. In how many different ways can he reach the top?
- What is the parity of the 20th term of the Virahāṅka sequence?
- Identify the statements that are true.
 - The expression $4m - 1$ always gives odd numbers.
 - All even numbers can be expressed as $6j - 4$.
 - Both expressions $2p + 1$ and $2q - 1$ describe all odd numbers.
 - The expression $2f + 3$ gives both even and odd numbers.
- Solve this cryptarithm:

$$\begin{array}{r}
 \text{UT} \\
 + \text{TA} \\
 \hline
 \text{TAT}
 \end{array}$$

Exercise

Multiple Choice Questions

LEVEL - 1

Parity of Numbers and Expressions

- If a 5×5 square grid has only odd numbers, what is the parity of the sum?
(a) Always odd
(b) Always even
(c) Can't say
(d) Always prime (NCERT Page 131)
- Which sum is not possible with 3 different even numbers?
(a) 12 (b) 18
(c) 15 (d) 14 (Page 131) **U**
- Choose the expression which always result in an even number for any value of x .
(a) $x + x + 1$
(b) $3x + 2$
(c) $2x$
(d) None of these (Page 132)
- Odd $\times$ Even = ?
(a) Odd
(b) Even
(c) Depends on digits
(d) Can't say **U**
- A light bulb is OFF. It toggles every time a switch is pressed. After 25 presses, the bulb will be:
(a) ON (b) OFF
(c) Blinking (d) Can't say **An**
- A tile pattern follows black, white and red. First tile is black. What colour is tile 15?
(a) Black
(b) White
(c) Red
(d) Can't say

Magic Squares, Virahāṅka Numbers and Cryptarithms

- A 3×3 magic square has its bottom row as 4, 3, 8. What is its center number?
(a) 5 (b) 6
(c) 2 (d) 7 (Page 135) **U**
- In a 3×3 magic square (1-9), if all the numbers are increased by 2, the new magic sum will be :
(a) Remains same
(b) Increased by 6
(c) Increased by 18
(d) Increased by 9 (Page 136)
- How many even numbers are in the classic Chautisā Yantra?
(a) 8 (b) 10
(c) 12 (d) 16 (Page 137)
- Which pair could NOT be consecutive terms in a Virahāṅka sequence?
(a) 5, 8 (b) 8, 13
(c) 21, 35 (d) 34, 55 (Page 139) **An**
- Which number is not in Fibonacci series?
(a) 3 (b) 7
(c) 8 (d) 13 (Page 139) **U**
- What is the parity of the 16th term in the Virahāṅka sequence?
(a) Even (b) Odd
(c) Divisible by 3 (d) None of these (Page 139-141)
- A 9-beat verse has how many rhythmic patterns?
(a) 55 (b) 65
(c) 89 (d) 34 (Page 139-141)
- If a staircase has 7 steps, and you can take 1 or 2 steps at a time, by how many different ways can you climb it?
(a) 13
(b) 21
(c) 34
(d) 8 (Page 139-141) **Ev**

15. In the cryptarithm : ($A \neq B$)

$$\begin{array}{r} AB \\ + BA \\ \hline CC \end{array}$$

What is the likely value of C?

- (a) 9 (b) 1
(c) 0 (d) 2 (Page 143) **Ap**

LEVEL - 2

16. A student skips every 3rd number while counting. What is the 7th number they will say?

- (a) 9 (b) 10
(c) 12 (d) 13 **Ap**

17. Which term in the Virahāṅka sequence is closest to 100?

- (a) 89 (b) 144
(c) 55 (d) 121 **Ev**

18. Which of these always results in a multiple of 4?

- (a) $2(x + 2)$ (b) $4x + 2$
(c) $x + x + x + x$ (d) All of these **An**

19. If each number in a 5×5 magic square (1-25) is increased by 5, the new magic sum will:

- (a) Increase by 25 (b) Remain same
(c) Increase by 125 (d) Increase by 5

20. The sum of first 15 Virahāṅka numbers is:

- (a) 2582 (b) 2528
(c) 2852 (d) None of these **Ev**

21. In a 5×5 grid, the centre number is 13. What must be the magic sum if each row adds equally?

- (a) 65 (b) 60 (c) 75 (d) 70 **U**

22. How many even numbers are there between 1 and 31 (inclusive)?

- (a) 14 (b) 15 (c) 16 (d) 13

23. A bulb is toggled 30 times. If it flickers after each 12th times it toggled. What is its final state if it starts OFF?

- (a) ON (b) OFF
(c) Flickering (d) Can't say **An**

24. In a code, where $A = 1, B = 2, \dots, Z = 26$, the sum of the letters of the word "MATH" is:
(a) 42 (b) 45 (c) 49 (d) 52

25. In a modified Virahāṅka sequence, where the first two terms are 2 and 3, which of the following is true?

- (a) Every 3rd term is even
(b) The sequence will include 13
(c) It will never include any multiple of 5
(d) All terms will be odd

26. If you reverse the digits of a two-digit number and subtract it from the original, then which of the following is always true?

- (a) Result is divisible by 11
(b) Result is odd
(c) Result is even
(d) Result can be even or odd **Ev**

27. The expression $(x + 1)(x - 1)$ gives:

- (a) an odd number always
(b) an even number only if x is odd
(c) always divisible by 2
(d) none of these **An**

28. If the sum of 2 odd numbers and a number (x) is even, then (x) :

- (a) odd (b) even
(c) same
(d) Both (a) and (c) are possible

29. A wall tiles pattern goes Blue, Red, Green, Blue, Red, What colour is the 202nd tile?

- (a) Blue (b) Red
(c) Green (d) Can't say

30. A puzzle shows: $S + U + N = 30$
Each letter is a different odd number between 1 and 19. What is the value of N?

- (a) 5 (b) 7
(c) 9 (d) Can't say **U**

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

31. The parity of sum of 52th and 65th Virahāṅka number (Starting from 1,1) is

- (a) odd (b) even
(c) prime (d) can't determine **U**

32. In Chautisā Yantra, if all the numbers are increased by 1 and three numbers in a row are 17, 4 and 6, what must the fourth number be to complete the row?
(a) 11 (b) 10 (c) 9 (d) 7

33. Which of the following expression always has odd parity?
(a) $5x + 2$ (b) $x + 17$
(c) $6x + 1$ (d) None of these

Ap

34. If we add two even numbers and subtract the result from product of odd and even number, the parity of result obtained will always be
(a) odd (b) even
(c) prime (d) none of these

U

35. The centre value of a 3×3 magic square whose magic sum is 30, is
(a) 15 (b) 12 (c) 10 (d) 9

Ap

36. The parity of sum of first 15 Virahāṅka numbers is
(a) Even (b) Odd
(c) Can't say (d) None of these

37. $AB + AB + AB = CCC$, then the maximum possible sum of A, B and C is
(a) 10 (b) 11 (c) 12 (d) 13

Ev

38. What will be the unit digit of the 21st Fibonacci number starting from 1,1?
(a) 5 (b) 6 (c) 7 (d) 8

Ev

39. Given, $5A9 + 3B7 + 2C8 = 1114$, then the maximum value of C is
(a) 5 (b) 7
(c) 8 (d) None of these

40. If each number in a 3×3 magic square is replaced with its square, does the square still remain magic?
(a) Yes
(b) Only if centre is 5
(c) Only if all numbers are even
(d) No

U

Match the Following

In this section each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which, one is correct.

1. Match the following

List-I

List-II

- | | |
|--|-------|
| (P) Expression $7x$ is odd for $x =$ | 1. 3 |
| (Q) Number of odd numbers from 1 to 20 is | 2. 2 |
| (R) 6 th Fibonacci number (starting with 1, 1) is | 3. 10 |
| (S) Number of diagonals in 3×3 magic square is | 4. 8 |
| (a) P-1, Q-3, R-4, S-2 | |
| (b) P-1, Q-3, R-2, S-4 | |
| (c) P-3, Q-1, R-2, S-4 | |
| (d) P-3, Q-1, R-4, S-2 | |

2. Match the following :

List-I

List-II

- | | |
|--|-------|
| (P) 7 th virahāṅka number is | 1. 2 |
| (Q) Minimum number needed to replace to make sum all rows odd in a 2×2 grid of even numbers | 2. 3 |
| (R) Number of different possible 3×3 magic squares (1-9) is | 3. 21 |
| (S) In a cryptarithm, $A + BA = 16$, then value of A is | 4. 8 |
| (a) P-1, Q-3, R-4, S-2 | |
| (b) P-4, Q-1, R-3, S-2 | |
| (c) P-3, Q-1, R-2, S-4 | |
| (d) P-3, Q-1, R-4, S-2 | |

3. Match the following :

List-I

List-II

- | | |
|---|-------|
| (P) Total number of odd multiples of 3 from 1 to 35 is | 1. 21 |
| (Q) In Fibonacci : Sum of 5 th and 6 th term is | 2. 11 |
| (R) Centre of a 3×3 magic square is | 3. 6 |
| (S) If $ABC = 182$, then sum of digits of A, B, C is | 4. 5 |
| (a) P-1, Q-3, R-4, S-2 | |
| (b) P-4, Q-1, R-3, S-2 | |
| (c) P-3, Q-1, R-2, S-4 | |
| (d) P-3, Q-1, R-4, S-2 | |

Assertion & Reason Type

Directions : In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as

- (a) *If both assertion and reason are true and reason is the correct explanation of assertion.*
- (b) *If both assertion and reason are true but reason is not the correct explanation of assertion.*
- (c) *If assertion is true but reason is false.*
- (d) *If assertion is false but reason is true.*

1. **Assertion :** A 4×4 magic square made using only even numbers can have an odd magic sum.
Reason : Even numbers when added in even or odd count always results in even.
2. **Assertion :** A cryptarithm puzzle cannot have two distinct letters representing the same digit.
Reason : In cryptarithm, each unique letter must map to a unique digit from 0 to 9.
3. **Assertion :** The 21st term of the Virahāṅka sequence is odd.
Reason : Every third term of the Virahāṅka sequence is odd.
4. **Assertion :** A 2×3 (1-6) grid can be filled with 3 odd and 3 even numbers such that each row and column sum has a specific parity.
Reason : The parity of a sum depends on the number of odd numbers or even numbers involved.
5. **Assertion :** If Angaan climbs a 12-steps staircase with steps of 1 or 2, then the total number of ways he can do it equals the 11th Virahāṅka number.
Reason : The number of paths in such problems follows the same pattern as the Virahāṅka sequence.

6. **Assertion :** The sum of page numbers in 20 sheets (each having 2 pages) is always even.
Reason : The sum of any two consecutive integers is always even.
7. **Assertion :** In a 3×3 magic square with a magic sum of 0, not all entries can be zero if many be negative.
Reason : Any number repeated in all positions of a magic square can produce a zero sum.
8. **Assertion :** The expression $x(x + 1)$ always gives an even number for any integer value of x .
Reason : Any product of two consecutive integers is divisible by 2.
9. **Assertion :** In the number puzzle, where $AB + BA = 121$, A must be 6.
Reason : Reversing the digits in a two-digit number and adding to the original always gives a multiple of 11.
10. **Assertion :** The expression $4x + 1$ always gives an odd number.
Reason : Product of an even and odd number is always an even number.

Comprehension Type

PASSAGE-I : Ritu created a rhythmic dance, where each beat could last 1 or 2 seconds. She wanted to choreograph a routine that lasted exactly 7 seconds. She challenged her class to find out in how many different ways the rhythm could be arranged.

1. Which mathematical pattern fits Ritu's beat combinations?
(a) Multiplication table
(b) Magic square
(c) Virahāṅka (Fibonacci) sequence
(d) Prime numbers
2. How many different combinations exist for 7-seconds rhythm?
(a) 13 (b) 21 (c) 34 (d) 55

3. If Ritu increases the rhythm duration to 10 seconds, what will happen to the number of combinations?
- It stays the same
 - It decreases
 - It follows the 10th Virahāṅka number
 - It becomes a multiple of 10

PASSAGE-II : In her math notebook, Kiara wrote the numbers from 1 to 20 and started coloring the even numbers in blue and the odd numbers in red. She noticed that when two odd numbers were added, the result was always even. But when she added an even and an odd, the result was always odd.

She called this "parity pairing rule" and used it to solve tricky puzzles involving large sums.

- What is the result of Odd + Odd based on Kiara's rule?
 - Odd
 - Even
 - Prime
 - Can't say
- What is the parity (even or odd) of the sum of 3 odd numbers?
 - Always even
 - Always odd
 - Can't say
 - Always zero
- If a number is divisible by 2, then its parity is always
 - Prime
 - Even
 - Odd
 - Composite

PASSAGE-III : In the ancient temple of Parshvanath, a magical square called Chautisā Yantra lies carved in stone. The priests believe its power lies in the four corner numbers. A scholar once found that no matter how the square was rotated, the corner sum remained constant and equal to the total magic sum. He tried replacing the corner numbers with higher values, and found the magic sum increased evenly.

- What is the sum of corner numbers in a Chautisā Yantra?
 - 34
 - 64
 - Equal to the sum of all numbers from 1 to 16
 - None of these

- If each corner number is increased by 1, by how much will the corner sum increase?
 - 1
 - 4
 - 16
 - 2
- If each number in the Chautisā Yantra is multiplied by '2', then the new magic sum will be
 - 24
 - 34
 - 42
 - 68

Subjective Problems

Very Short Answer Type

- What is the parity of the number 177?
- Write the next two terms in the Virahāṅka sequence starting from 5, 8.
- What will be the result (odd or even) of multiplying an odd number with an even number?
- If the center number of a 3×3 magic square is 7, then what is the magic sum?
- Which letter in the word 'TOY' corresponds to the number 20 if $A = 1$, $B = 2$, $C = 3$, ... coding?
- What type of number (odd/even) is always produced by the expression $x(x - 1)$?
- State whether the expression $3x + 1$ is even or odd when $x = 4$.
- Is the number 21 a part of the Virahāṅka sequence? Justify.
- Create an expression which always results in an odd number for any value of x .
- Find 15th even and odd numbers.

Short Answer Type

- Is the product of any three consecutive natural numbers always divisible by 6? Explain briefly.
- A student says on skipping every 3rd number, the 7th number is 13, by what number did he start from?
- A class is divided into groups of 5. If there are 19 such groups, then find the total number of students odd or even? Justify.

Give a real-life situation where a pattern similar to the Fibonacci (Virahāṅka) sequence can be observed.

In a cryptarithm, the word "SUN" represents a 3-digit number where each letter stands for a different digit. If $S = 9$ and $U = 2$, what is the smallest possible value of N for the number to be divisible by 5?

Find the values of the letters in each of the following and give reasons for the steps involved.

$$\begin{array}{r} 3 \ A \\ + 2 \ 5 \\ \hline B \ 2 \end{array}$$

- A pattern goes like this: 2, 4, 8, 16, What type of sequence is this, and what is the next number?
- A magician asks an audience member to pick any 3 odd numbers. He adds them up and claims the result is even. Is this true? Justify your answer using parity.
- Create a 3×4 grid by using 6 odd and 6 even numbers such that all rows has even sum.
- In nature, the number of petals in flowers often follows the Fibonacci (Virahāṅka) sequence. If a flower has 13 petals, what are the two previous likely numbers of petals in this pattern?

Long Answer Type

- Create a 3×3 grid contains only odd numbers. Can the sum of each row be even? Explain your reasoning and give an example.
- Create your own version of the Fibonacci sequence where the first two numbers are 4 and 6. Write the first 8 terms and explain how it is similar and different from the standard sequence.
- Explore the Chautīsā Yantra (ancient 4×4 magic square with a sum of 34). Explain how rotation or replacement of corner values changes or preserves the magic sum.

- Form two expressions in which one results in odd number and one in even number for any value of x . Try it for $x = 1, 2, 3$, and explain why it always results in an even or odd number.
- Solve the following cryptarithm :
 $OP + OP + OP + OP = QO$ or $4 \times OP = QO$



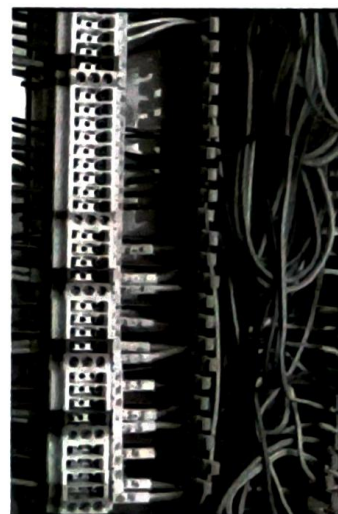
Integer/Numerical Value Type

- Calculate the difference between the 15th and 10th term of the Fibonacci sequence starting from 1, 1.
- If the magic sum of a 3×3 magic square is 36, find the centre number n .
- If a 4×4 magic square (Chautisa Yantra) has a magic sum of 34, what should be the sum of numbers in the corners?
- A 3-digit number ABC is such that $A = C$, $B = 2$ and the number is divisible by 11. What is the sum of A and C ?
- If $TA + AT = 99$, then find the maximum value of A . (Where $A, T \neq 0$)



Case Based Questions

Case I : Arya is building a circuit where each wire carries a signal, coded by a number. She realizes the signal behaves according on its parity.



Based on the above information, answer the following questions.

CFQs

1. If two wires carry odd numbers, the result of combining them is always
(a) Odd (b) Even
(c) Prime (d) Negative
2. If a wire carries the expression $8x + 5$. What type of number does it generate?
(a) Always odd
(b) Always even
(c) Alternates (Even – Odd)
(d) Always divisible by 3
3. Which pair of numbers will give an odd result when added?
(a) $5 + 7$ (b) $6 + 6$
(c) $9 + 2$ (d) $3 + 3$
4. Arya gets the sum of four odd numbers is:
(a) Odd (b) Even
(c) Always 0 (d) Always a square
5. A switch activates only when the product is odd. Which pair will not activate it?
(a) 3 and 5 (b) 7 and 9
(c) 4 and 5 (d) 11 and 13

Case II : Rina is breeding rabbits by following a modified Virahāṅka (Fibonacci) rule: each month, the number of pairs is the sum of the previous two months. She starts with 3 and 5 pairs.



Based on the above information, answer the following questions.

CFQs

1. What is the number of pairs in the 4th month?
(a) 13 (b) 10 (c) 11 (d) 8
2. Which of the following is a correct continuation of this modified sequence?
(a) 3, 5, 8, 12 (b) 3, 5, 8, 13
(c) 3, 5, 7, 12 (d) 3, 5, 9, 14
3. In the 6th month, the number of pairs is
(a) 34 (b) 21 (c) 24 (d) 26
4. In which month, Rina will have 89 pairs?
(a) 5th (b) 8th
(c) 12th (d) None of these
5. If Rina sold all pairs after 10 months for ₹15 per pair, then the amount collected by her will be
(a) ₹ 3295 (b) ₹ 3485
(c) ₹ 3490 (d) ₹ 3495